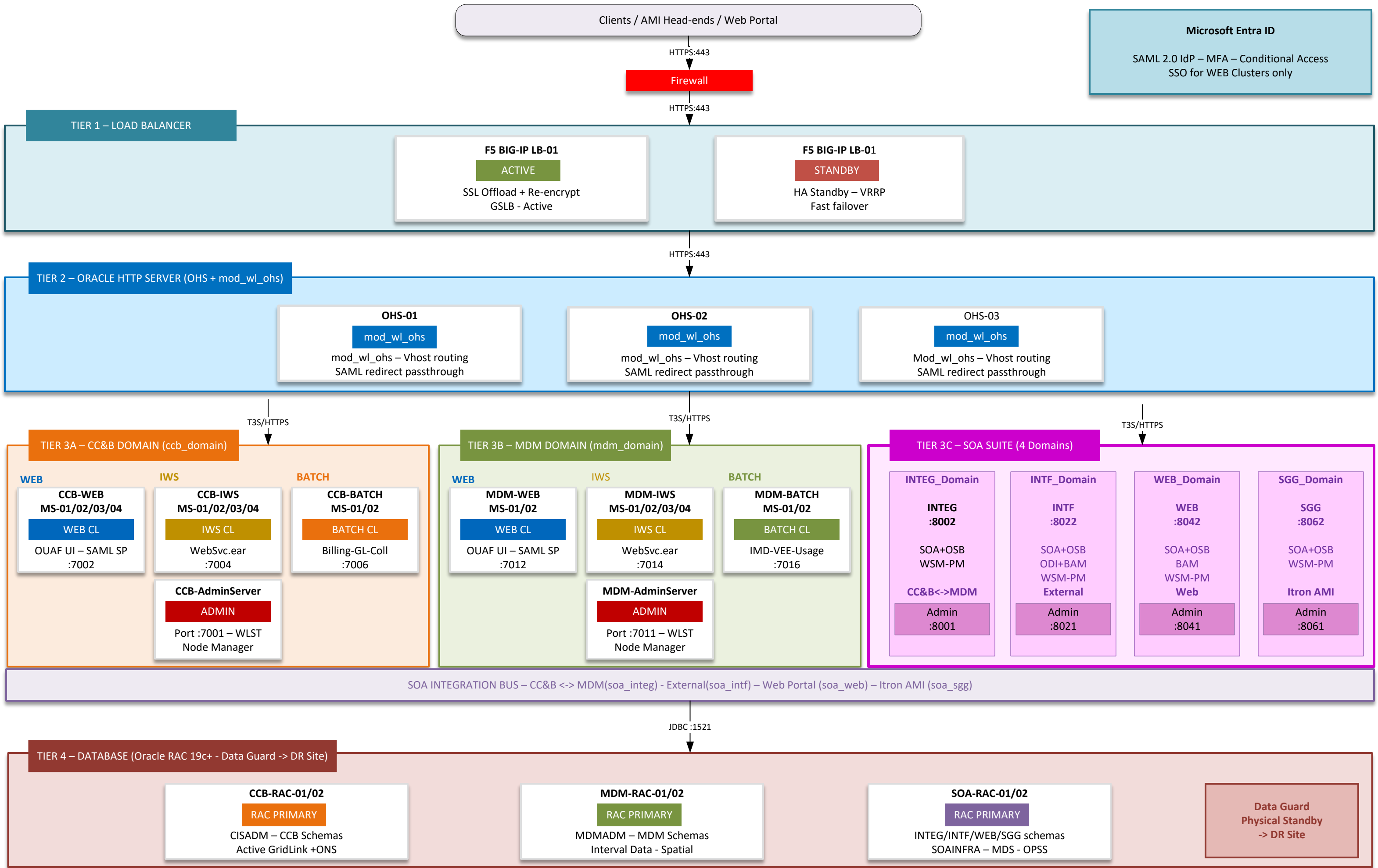




AdminServer – CCB-AdminServer (Port :7001 – WLS Console – WLST – OUAF Admin UI – iwsdeploy target – Node Manager on dedicated VM)

CCB-WEB Cluster

Purpose: OUAF Web UI – ADF
Entra ID SAML SP
Session replication ON
OHS routed via mod_wl_ohs

OHS route: /ccb -> CCB-WEB

Oracle Linux VM: ccb-web-vm-01

CCB-WEB-MS-01 :7002
WEB

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/ccb_domain

Oracle Linux VM: ccb-web-vm-02

CCB-WEB-MS-02 :7002
WEB

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/ccb_domain

Oracle Linux VM: ccb-web-vm-03

CCB-WEB-MS-03 :7002
WEB

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/ccb_domain

Oracle Linux VM: ccb-web-vm-04

CCB-WEB-MS-04 :7002
WEB

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/ccb_domain

CCB-IWS Cluster

Purpose: WebServices.ear
SOAP/REST APIs
WS-Security
iwsdeploy target
OHS routed via mod_wl_ohs

OHS route: /ccb-iws -> CCB-IWS

Oracle Linux VM: ccb-iws-vm-01

CCB-IWS-MS-01 :7004
IWS

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/ccb_domain

Oracle Linux VM: ccb-iws-vm-02

CCB-IWS-MS-02 :7004
IWS

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/ccb_domain

Oracle Linux VM: ccb-iws-vm-03

CCB-IWS-MS-03 :7004
IWS

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/ccb_domain

Oracle Linux VM: ccb-iws-vm-04

CCB-IWS-MS-04 :7004
IWS

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/ccb_domain

CCB-BATCH Cluster

Purpose: Bill Generation – GL
Payment Processing
Collections – Rate Eng.
NOT routed vis OHS

OHS route: Internal only

Oracle Linux VM: ccb-batch-vm-01

CCB-BATCH-MS-01 :7006
BATCH

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/ccb_domain

Oracle Linux VM: ccb-batch-vm-01

CCB-BATCH-MS-01 :7006
BATCH

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/ccb_domain

MDM-WEB Cluster

Purpose: OUAF MDM UI
360 Device View
Entra ID SAML SP
Session replication ON

OHS route: /ccb -> CCB-WEB

Oracle Linux VM: mdm-web-vm-01

MDM-WEB-MS-01 :7012

WEB

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/mdm_domain

Oracle Linux VM: mdm-web-vm-02

MDM-WEB-MS-02 :7012

WEB

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/mdm_domain

MDM-IWS Cluster

Purpose: MDM WebServices.ear
SOA-SGG integration
SOA-INTEG usage feeds
WS-Security

OHS route: /ccb-iws -> CCB-IWS

Oracle Linux VM: mdm-iws-vm-01

MDM-IWS-MS-01 :7014

IWS

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/mdm_domain

Oracle Linux VM: mdm-iws-vm-01

MDM-IWS-MS-01 :7014

IWS

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/mdm_domain

Oracle Linux VM: mdm-iws-vm-03

MDM-IWS-MS-03 :7014

IWS

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/mdm_domain

Oracle Linux VM: mdm-iws-vm-04

MDM-IWS-MS-04 :7014

IWS

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/mdm_domain

MDM-BATCH Cluster

Purpose: IMD Ingest – VEE Engine
Usage Calculation
Exception Management
SGG JMS queue consumer

OHS route: Internal only

Oracle Linux VM: mdm-batch-vm-01

MDM-BATCH-MS-01 :7016

BATCH

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/mdm_domain

Oracle Linux VM: mdm-batch-vm-01

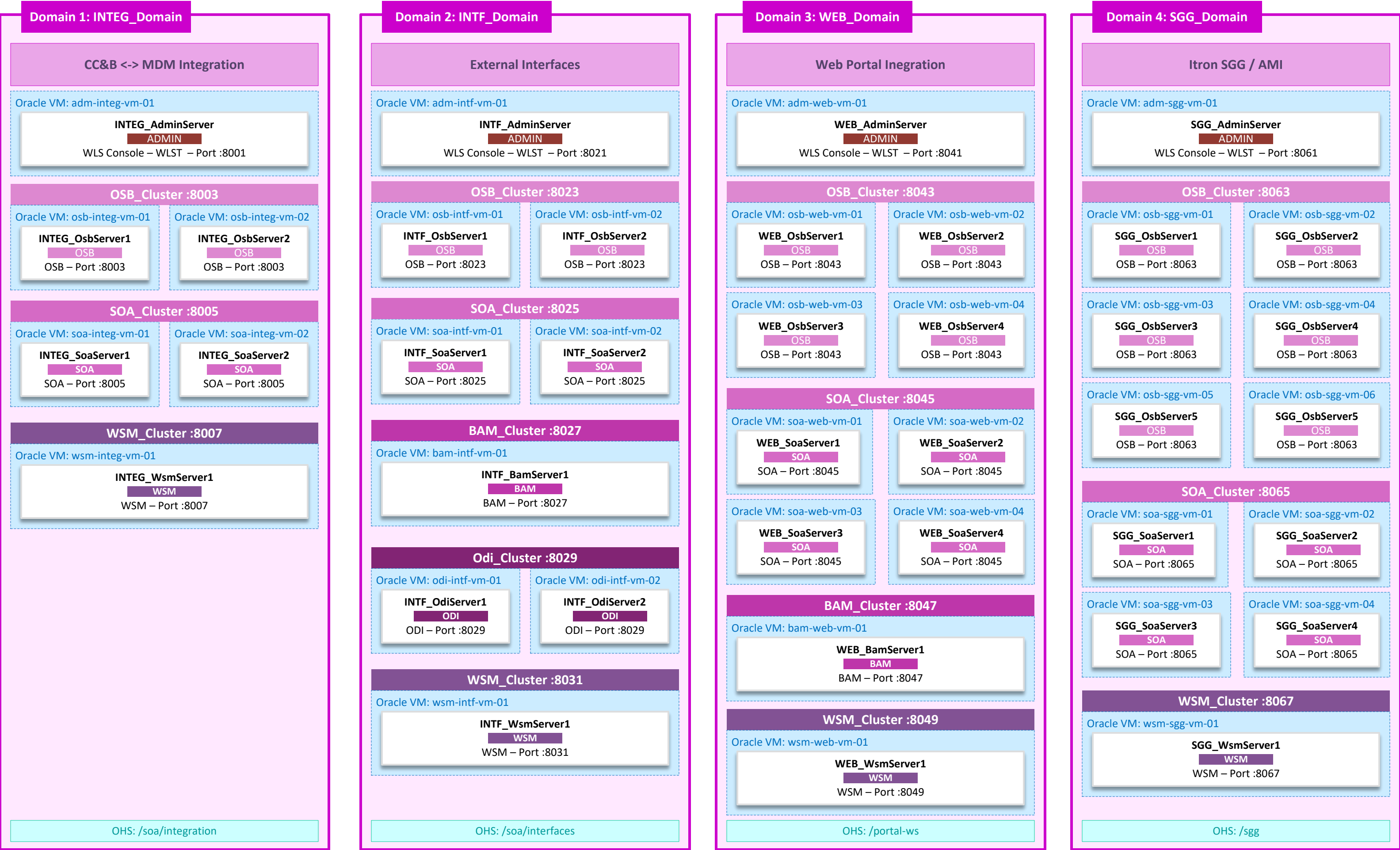
MDM-BATCH-MS-01 :7016

BATCH

Node Manager – auto-restart
JDK17 – WLS 14.1.1

Domain home: /u01/oracle/config/mdm_domain

← SOA-SGG (Itron) feeds IMD to this cluster's JMS queue



Each domain has fully independent lifecycle – patching, composite deployment, and restarts in one domain have zero impact on others. WSM-PM is local to each domain (no shared WSM).

CCB-RAC

CCB-RAC-01

PRIMARY

RAC Node – CRS / ASM

SCAN Listener – ONS :6200

CCB-RAC-02

PRIMARY

RAC Node – CRS / ASM

SCAN Listener – ONS :6200

SCHEMAS

CISADM (CC&B app)

CISREAD (reporting)

CISSYS (OUAF framework)

DB SERVICES

CCB_OLTP

Both RAC nodes – WEB+IWS

CCB_BATCH

Preferred: RAC-02 - Batch

SCAN: ccb-scan.duqlight.com

Est. size: 10 TB+

Data Guard → Physical Standby at DR site

MDM-RAC

MDM-RAC-01

PRIMARY

RAC Node – CRS / ASM

SCAN Listener – ONS :6200

MDM-RAC-02

PRIMARY

RAC Node – CRS / ASM

SCAN Listener – ONS :6200

SCHEMAS

MDMADM (CC&B app)

MDMREAD (reporting)

Oracle Spatial (meter loc)

DB SERVICES

MDM_ONLINE

Both RAC nodes – WEB+IWS

MDM_IMD

Preferred: RAC-02 – Batch IMD

SCAN: mdm-scan.duqlight.com

Est. size: 80 TB+

Data Guard → Physical Standby at DR site

SOA-RAC

CCB-RAC-01

PRIMARY

RAC Node – CRS / ASM

SCAN Listener – ONS :6200

CCB-RAC-02

PRIMARY

RAC Node – CRS / ASM

SCAN Listener – ONS :6200

SCHEMAS

INTEG/INTF/WEB/SGG_SOAINFRA

INTEG/INTF/WEB/SGG_MDS

INTEG/INTF/WEB/SGG_OPSS+IAU+STB

DB SERVICES

SOA_INTEG/ITNF/WEB/SGG

Per-domain services

MDS_*

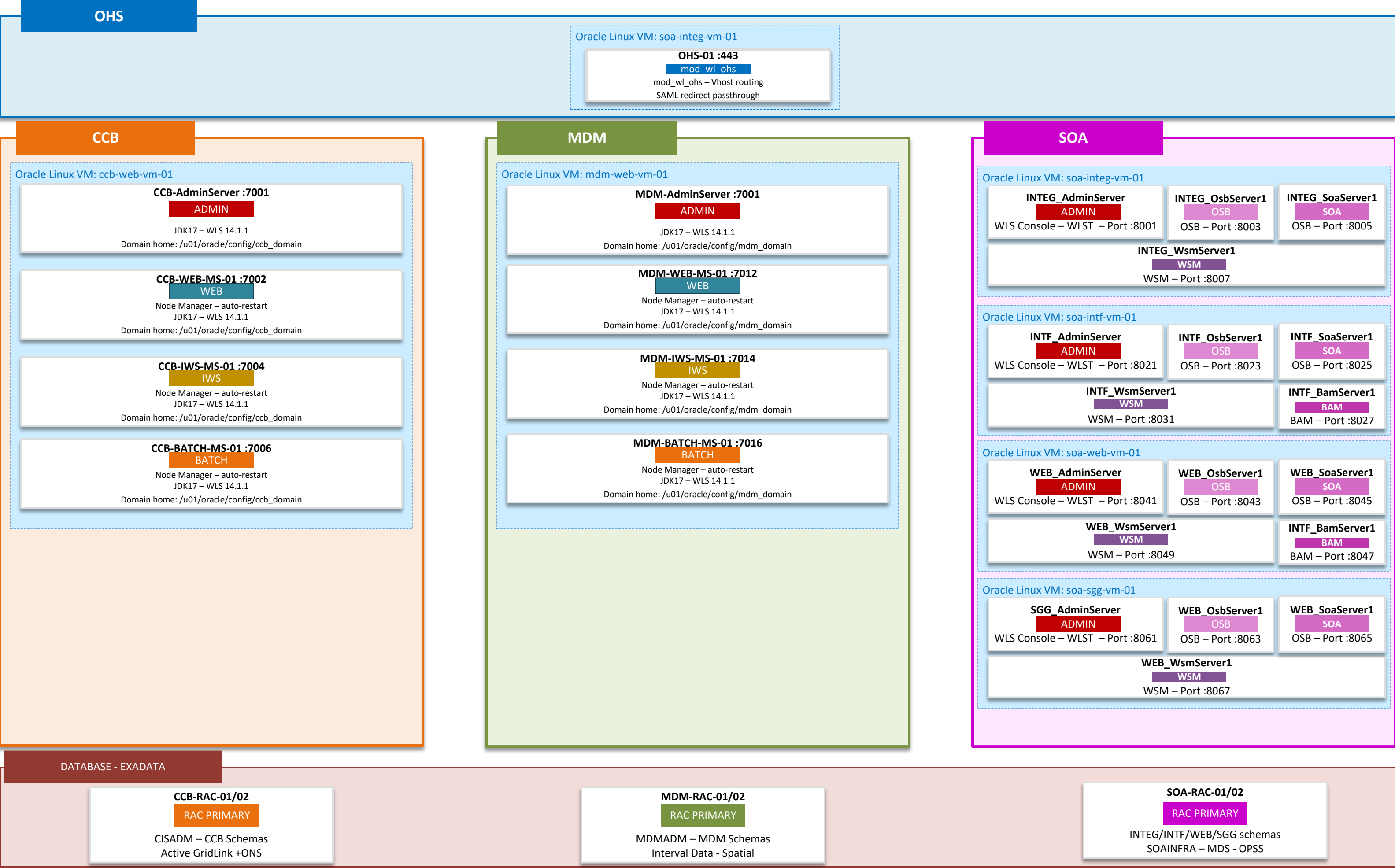
Metadata per domain

SCAN: soa-scan.duqlight.com

Est. size: 3 TB+

Data Guard → Physical Standby at DR site

DATA GUARD CONFIGURATION – ALL THREE CLUSTERS					
Protection Mode	Standby Type	Fast-Start Failover	App Continuity	RPO Target	GDS
Maximum Availability (SYNC+AFFIRM)	Physical Standby (2-node RAC at DR site)	Enabled – Observer at 3 rd location – 30s threshold	Active GridLink – transparent replay on failover	< 30 seconds	Global Data Services for client failover



DATABASE - EXADATA

CCB-RAC-01/02

RAC PRIMARY

CISADM – CCB Schemas

Active GridLink +ONS

MDM-RAC-01/02

RAC PRIMARY

MDMADM – MDM Schemas

Interval Data - Spatial

SOA-RAC-01/02

RAC PRIMARY

INTEG/INTF/WEB/SGG schemas

SOAINFRA – MDS - OPSS

Greentree – Primary Production

Dell PowerEdge R760
Oracle Linux KVM

Dell PowerEdge R760
Oracle Linux KVM

Dell PowerEdge R760
Oracle Linux KVM

Dell PowerEdge R760
Oracle Linux KVM

Dell PowerEdge R760
Oracle Linux KVM

Dell PowerEdge R760
Oracle Linux KVM

Everpure (formerly Purestorage) Flasharray

Penn Hills – Non-Production

Dell PowerEdge R760
Oracle Linux KVM

Dell PowerEdge R760
Oracle Linux KVM

Dell PowerEdge R760
Oracle Linux KVM

Dell PowerEdge R760
Oracle Linux KVM

Dell PowerEdge R760
Oracle Linux KVM

Dell PowerEdge R760
Oracle Linux KVM

Dell PowerEdge R760
Oracle Linux KVM

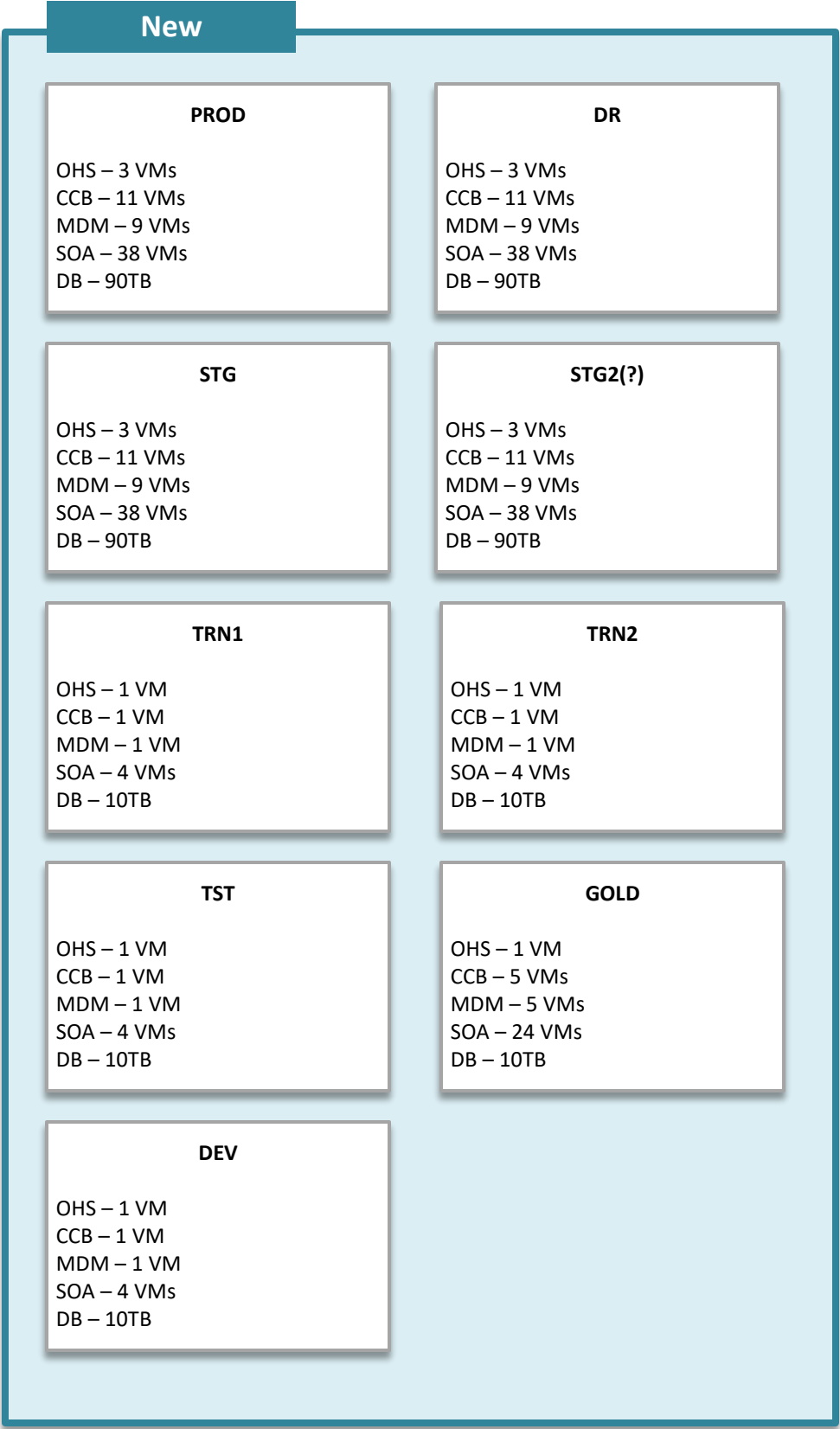
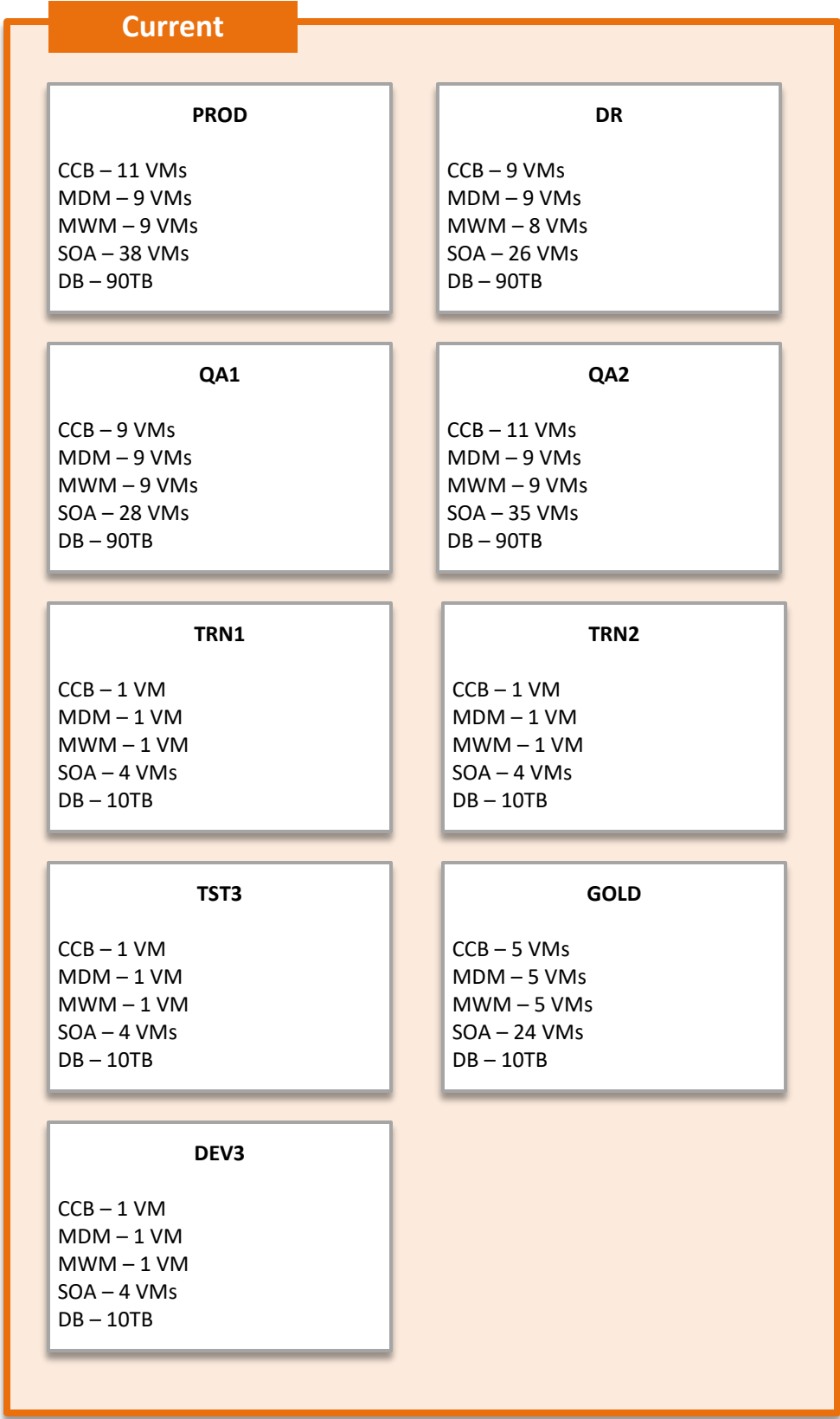
Dell PowerEdge R760
Oracle Linux KVM

Dell PowerEdge R760
Oracle Linux KVM

Dell PowerEdge R760
Oracle Linux KVM

Dell PowerEdge R760
Oracle Linux KVM

Everpure (formerly Purestorage) Flasharray



Greentree

PROD - Current

CCB – 11 VMs
MDM – 9 VMs
MWM – 9 VMs
SOA – 38 VMs
DB – 90TB

PROD – New

OHS – 3 VMs
CCB – 11 VMs
MDM – 9 VMs
SOA – 38 VMs
DB – 90TB

Penn Hills

DR - Current

CCB – 9 VMs
MDM – 9 VMs
MWM – 8 VMs
SOA – 26 VMs
DB – 90TB

DR - New

OHS – 3 VMs
CCB – 11 VMs
MDM – 9 VMs
SOA – 38 VMs
DB – 90TB

QA1 - Current

CCB – 9 VMs
MDM – 9 VMs
MWM – 9 VMs
SOA – 28 VMs
DB – 90TB

QA2 - Current

CCB – 11 VMs
MDM – 9 VMs
MWM – 9 VMs
SOA – 35 VMs
DB – 90TB

STG - New

OHS – 3 VMs
CCB – 11 VMs
MDM – 9 VMs
SOA – 38 VMs
DB – 90TB

STG2(?) - New

OHS – 3 VMs
CCB – 11 VMs
MDM – 9 VMs
SOA – 38 VMs
DB – 90TB

TRN1 - Current

CCB – 1 VM
MDM – 1 VM
MWM – 1 VM
SOA – 4 VMs
DB – 10TB

TRN2 - Current

CCB – 1 VM
MDM – 1 VM
MWM – 1 VM
SOA – 4 VMs
DB – 10TB

TRN1 - New

OHS – 1 VM
CCB – 1 VM
MDM – 1 VM
SOA – 4 VMs
DB – 10TB

TRN2 - New

OHS – 1 VM
CCB – 1 VM
MDM – 1 VM
SOA – 4 VMs
DB – 10TB

TST3 - Current

CCB – 1 VM
MDM – 1 VM
MWM – 1 VM
SOA – 4 VMs
DB – 10TB

GOLD - Current

CCB – 5 VMs
MDM – 5 VMs
MWM – 5 VMs
SOA – 24 VMs
DB – 10TB

TST - New

OHS – 1 VM
CCB – 1 VM
MDM – 1 VM
SOA – 4 VMs
DB – 10TB

GOLD – New

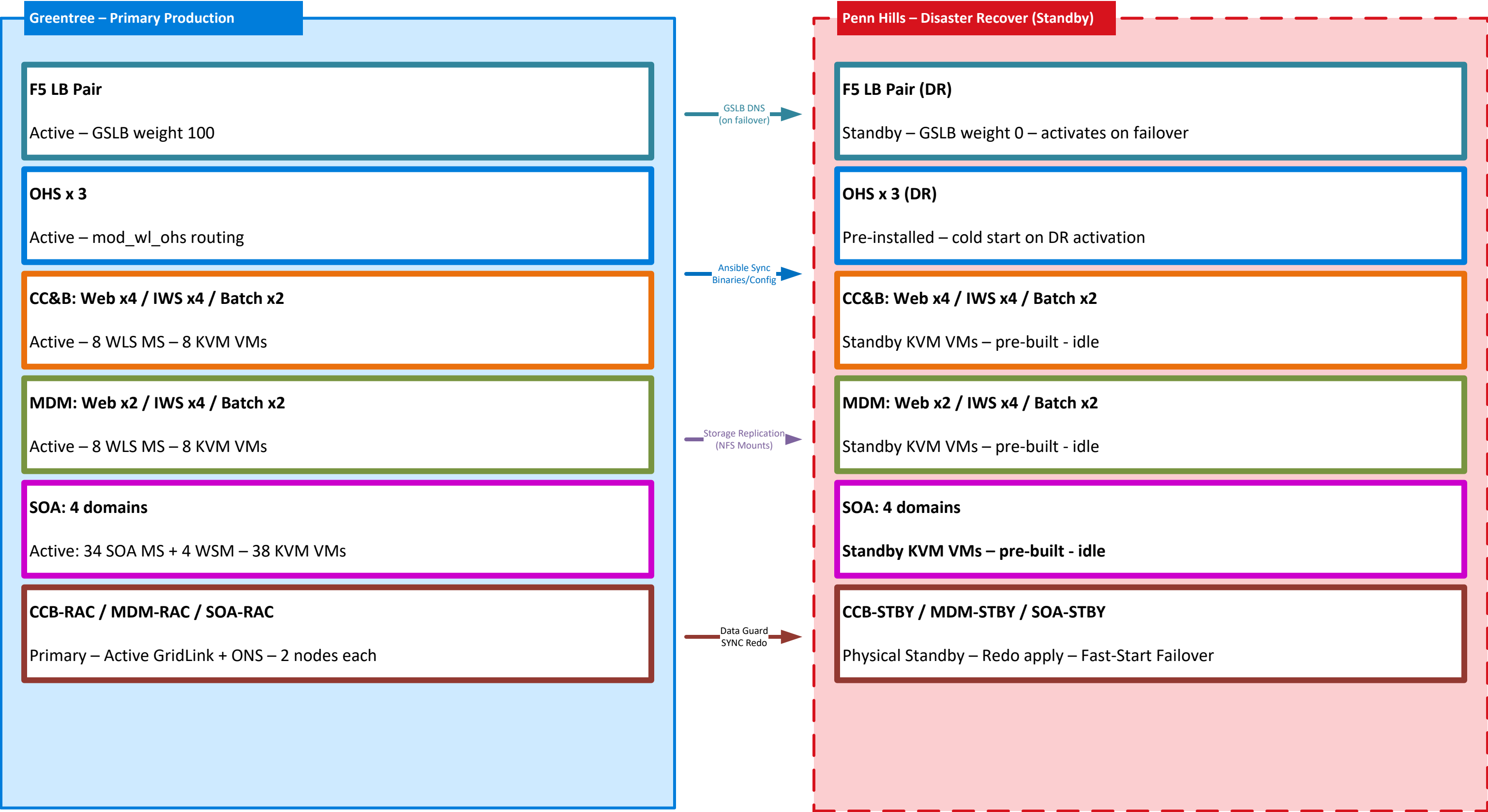
OHS – 1 VM
CCB – 5 VMs
MDM – 5 VMs
SOA – 24 VMs
DB – 10TB

DEV3 - Current

CCB – 1 VM
MDM – 1 VM
MWM – 1 VM
SOA – 4 VMs
DB – 10TB

DEV - New

OHS – 1 VM
CCB – 1 VM
MDM – 1 VM
SOA – 4 VMs
DB – 10TB



DR Runbook (Est. RTO ~30-45 min):

Entra ID – Identity Provider (IdP)
SAML 2.0 – MFA – Conditional Access – tenant-level HA

CCB-Web-Prod

ACS: <https://ccb.duqlight.com/saml2/sp/acs/post>

MDM-Web-Prod

ACS: <https://mdm.duqlight.com/saml2/sp/acs/post>

SOA-Console-Prod

ACS: <https://soa-admin.duqlight.com/saml2/sp/acs/post>

SAML 2.0 SP-Initiated Flow:

1. User accesses ccb.duqlight.com → LB → OHS → CCB-WEB Cluster
2. WLS generates SAMLAuthnRequest → 302 redirect to Entra ID
3. Entra ID authenticates user (+ MFA) → issues SAML Response
4. Browser POSTs to ACS URL → WLS SAML Assertion Provider validates
5. WLS establishes session → OUAF loads user context from NameID (email)

SAML scope: WEB clusters only (CCB-WEB, MDM-WEB)
IWS clusters → WS-Security (service accounts). Batch → no interactive SS).
Entra ID metadata: remove <RoleDescriptor> tag before WLS import.

